

Solution

Let

$$N = 2025^2 = 4100625$$

be the number of cells of the board. A cell is *grey* if it has never been painted, *lime* if its final colour is lime and *scarlet* otherwise. Linda moves first, then the players alternate.

0.0.1 a) The game finishes after finitely many moves

Denote by g_t the number of grey cells after the t -th move of **Sue** (the only player who can change the number of grey cells). When Sue makes a move she must choose a 2×2 block containing a grey cell; all four cells of the block become scarlet. Hence at least one grey cell disappears, i.e.

$$g_{t+1} \leq g_t - 1 \quad \text{for every Sue-move.}$$

Initially $g_0 = N$. Because g_t is a non-negative integer which decreases by at least one at each of Sue's turns, after at most N Sue moves we have $g_t = 0$. As soon as there is no grey cell the game stops (the rule of the statement). Consequently the game ends after a finite number of moves.

0.0.2 b) The number of lime cells when both players play optimally

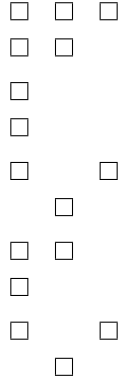
1. A useful decomposition of the board A **cross** (or **plus-pentomino**) is the set of five cells

$$C(i, j) = \{(i, j), (i-1, j), (i+1, j), (i, j-1), (i, j+1)\},$$

the centre together with the four orthogonal neighbours (all indices are taken inside the board). A cross contains exactly five cells.

Lemma 1. The 2025×2025 board can be partitioned into disjoint crosses.

Proof. Since $2025 = 5 \cdot 405$, the board is the disjoint union of 5×5 squares. A 5×5 square can be tiled by five crosses; for example



covers all 25 cells of the square, the five drawn figures being crosses. Repeating this tiling in each of the 405^2 squares gives a tiling of the whole board by $N/5$ disjoint crosses.

2. At most one lime cell per cross

Lemma 2. In a terminal position (no grey cells left) each cross contains at most one lime cell.

Proof. Consider a particular cross C .

If the centre of C is lime. When a cell is painted lime, all its orthogonal neighbours must already be non-grey; otherwise later Sue could select a block containing this cell and a still grey neighbour and would recolour the centre scarlet. Thus the four arm cells of the cross are non-grey, and a non-grey cell can never be lime in the final position (otherwise a later Sue move would have turned it scarlet). Hence none of the four arms can be lime.

If an arm of C is lime. Suppose, say, $(i-1, j)$ is lime. Its own orthogonal neighbours are the centre (i, j) and three cells lying outside the cross. The centre therefore must already be non-grey, so it cannot be lime. The three remaining arms of the cross are orthogonal neighbours of this arm as well; consequently they are also non-grey and cannot be lime.

In both cases the cross contains at most the one cell that is lime.

3. An upper bound for the number of lime cells

From Lemma 1 the board is the disjoint union of $N/5$ crosses, and from Lemma 2 each of them contributes at most one lime cell. Therefore

$$\boxed{\text{final number of lime cells} \leq N/5}$$

4. A lower bound – Linda can force $N/5$ lime cells

The inequality (1) will be sharp. Linda follows the strategy described below; Sue plays arbitrarily.

Strategy of Linda.

1. The crosses are numbered arbitrarily.

2. She processes the crosses one after another. While processing a given cross she never paints any cell belonging to a later cross.

For the cross that is being processed she first makes sure that its four arm cells become scarlet (this is possible because at the moment she starts the next cross at least four grey cells are still present; Sue can paint a 2×2 block consisting entirely of grey cells, thereby turning four of them scarlet). When the four arms are scarlet, Linda paints the centre cell of the cross lime. The centre will never be recoloured, because every 2×2 block that contains it also contains one of its arms, which is already scarlet; consequently such a block contains no grey cell and is no longer a legal move for Sue.

3. She repeats the procedure for the next cross.

Why the strategy works.

- In a cross the four arms are turned scarlet by four successive moves of Sue, each of which removes exactly four new grey cells. Hence while a cross is being processed exactly four of Sue's moves and exactly one of Linda's moves are used.
- The board contains $N/5$ crosses, therefore the whole game lasts exactly $N/5$ rounds (a round = Linda's move followed by Sue's move). After the last round no grey cell remains, so the game ends immediately after Sue's move of that round.
- Linda has painted the centre of every cross lime, i.e. she has produced exactly one lime cell per cross. Consequently the final board contains exactly $N/5$ lime cells.

Thus

$$\boxed{\text{final number of lime cells} \geq N/5}$$

5. End of the game is always a Sue-move If after Linda's move there were at most four grey cells left, Sue could choose a 2×2 block that contains all of them and would end the game on her turn. Consequently, with optimal play the game always terminates after a Sue move; the numbers of moves of the two players are equal. Hence the number of Linda moves equals the number of Sue moves, and the quantity proved in (2) is indeed the number of lime cells at the end.

6. Conclusion From (1) and (2) we obtain that under optimal play the number of lime cells is forced to be exactly

$$\frac{N}{5} = \frac{2025^2}{5} = 820\,125.$$

Answer.

a) The game always ends after finitely many moves (the number of grey cells strictly decreases after every move of Sue).

b) When both players use optimal strategies, exactly $\frac{2025^2}{5} = 820\,125$ cells are lime in the final position.